Artificial Intelligence Software, Lecture 3

AI Paper Pipeline: Writing, Reading, and Reviewing with AI

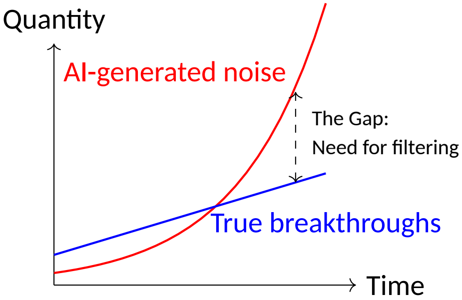
MIPT
Center for Cognitive Modeling

Outline

Introduction: The Researcher in the AI Era

- **Role Shift:** From generating text to generating meaning. AI is a tool, not the author.
- **Knowledge vs. Data:** AI models reflect existing ideas. New knowledge comes from researchers.
- **The Noise Problem:**
  - Easier to generate text → Flood of low-quality papers.
  - Speed vs. Quality trade-off.
- **Value of True Research:** High-quality, reproducible work stands out more than ever.

Publish or Perish: New Reality



- **Reproducibility** is the new competitive advantage.
- **Open Code** is a sign of respect.
- **Transparency** builds trust.

Responsibility and Reputation

- **Author's Responsibility:**
  - Setting the research problem.
  - Conducting experiments.
  - Interpreting results.
  - Ethics and safety.
- **Reputation:** AI can write, but it cannot own the results.
- **Expertise:** You are the navigator in the information noise.

Paper Lifecycle Overview

The Process Levels

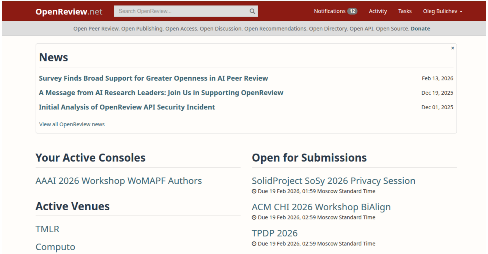
1. **Researcher's Work:**
   - Idea generation & Experiments.
   - Drafting (iterative process).
2. **Collaborative Work:**
   - Internal review (Colleagues).
   - AI pre-review (sanity check).
3. **Community Work:**
   - Submission (Conferences, Workshops, Journals).
   - External Review & Rebuttal.

Preparation for Submission

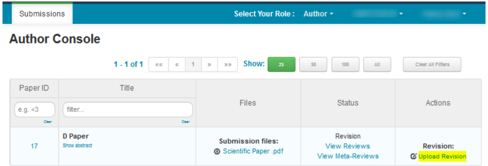
- **Deadlines:** Planning is key (Abstract deadline vs Full paper deadline).
- **Venue Selection:** ICLR, NeurIPS, ICML, CVPR, AAAI, etc.
- **Formal Requirements:**
  - Page limits (Strict!).
  - Templates (LaTeX).
  - Anonymity (Double-blind review).

Submission Systems

OpenReview.net (ICLR, NeurIPS)



Microsoft CMT (CVPR, ICCV)



Workshops and Journals

Workshops:

- Work in progress.
- Networking & Feedback.
- Specific narrow topics.
- Less strict than main conference.

Journals:

- Archival value.
- Longer review cycle (months/years).
- More detailed description.

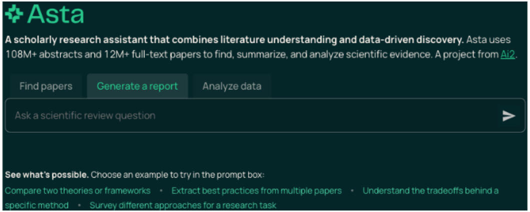
Reference Materials

- Conference websites (NeurIPS, ICLR, etc.) for requirements.
- OpenReview.net
- Microsoft CMT

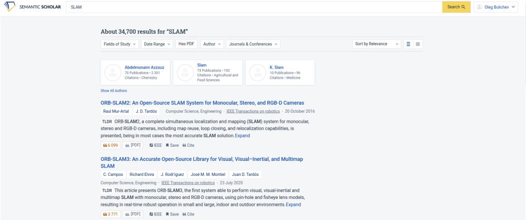
Searching for Literature

- **Initial Search:**
  - **Semantic Scholar:** standard for AI papers.
  - **Asta (Ai2):** emerging tool.
  - **Deep Research Agents:** perplexity, etc.
  - **Scholar inbox:** AI-powered literature search.
- Don’t just rely on Google. Use academic-specific engines.

Apps for Literature Search



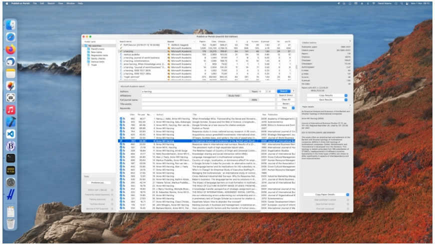
Asta (AI2)



Semantic Scholar

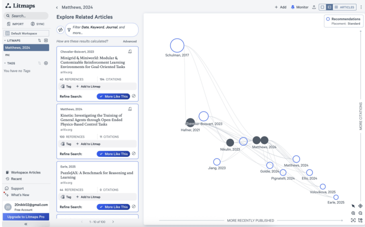
Publish or Perish (Application)

- **What is it?** A software program that retrieves and analyzes academic citations.
- **Key Features:**
  - Uses Google Scholar, Scopus, Web of Science.
  - Calculates h-index, g-index metrics.
  - Detailed citation analysis for authors or journals.
  - Helps identify "Must Read" classic papers by citation count.

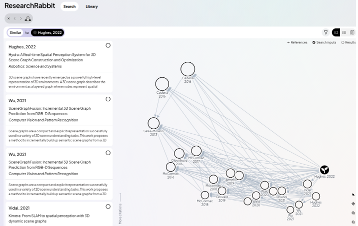


Visualization of Connections

- Build collections of papers.
- AI learns what you like and improves recommendations.
- Interactive visualization of author/paper networks.
- Infinite feed of related work.



(a) Litmap

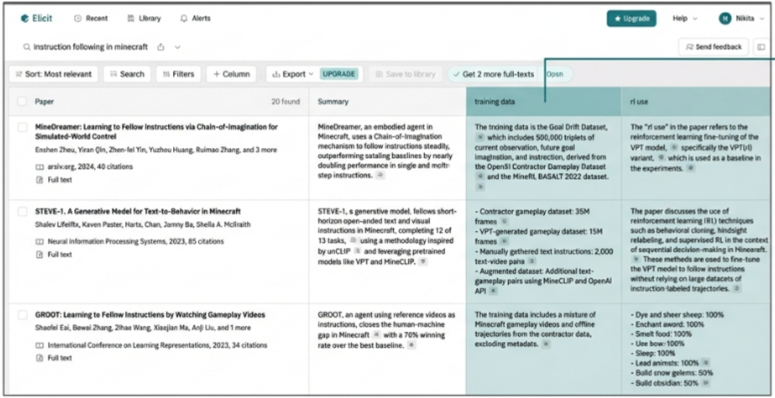


(b) ResearchRabbit

Structing & Information Extraction

- **Elicit:**
  - "Research Assistant".
  - Extract data from papers into a structured table.
  - "What is the NMAE score for X?" across 10 papers.
- **NotebookLM:** RAG system for synthesis.

Elicit



Automated Extraction Columns: Methodology, Dataset, Metrics, Limitations.

The MCP Pipeline (Advanced)

Automation with MCP (Model Context Protocol):

1. **Prompt:** Define strict search criteria.
2. **AI Agent + Google Scholar MCP:**
   - Agent queries Scholar directly via MCP.
   - Extracts Abstract & DOI.
   - Saves to Markdown.
3. **RAG Integration:** Feed md/pdfs to NotebookLM.
4. **Generation:** Synthesize Technical Report.

Reference Materials

- Google Scholar MCP Server
- ResearchRabbit
- Publish or Perish App
- Semantic Scholar
- Elicit.com
- Litmaps
- Perplexity
- Asta (Ai2)
- Scholar Inbox

Zotero: The Knowledge Hub

- **Why Zotero 8?** Modern UI, native speed, built-in PDF reader.
- **Browser Connector:** One-click save (PDF + Metadata) from ArXiv, IEEE, etc.
- **Built-in Features:**
  - Automatic file renaming based on parent metadata.
  - Extracting annotations to notes directly.
- **Note Templates:** Configurable templates regarding your workflow.

Essential Plugins

- **Better BibTeX:** Generates stable Citation Keys (e.g., bulichev2024ais) for LaTeX/Overleaf.
- **Zotero SciPDF:** Automatically downloads PDFs via Sci-Hub if paywalled.
- **Better Notes:** Advanced knowledge management, markdown support, and bi-directional linking within Zotero.

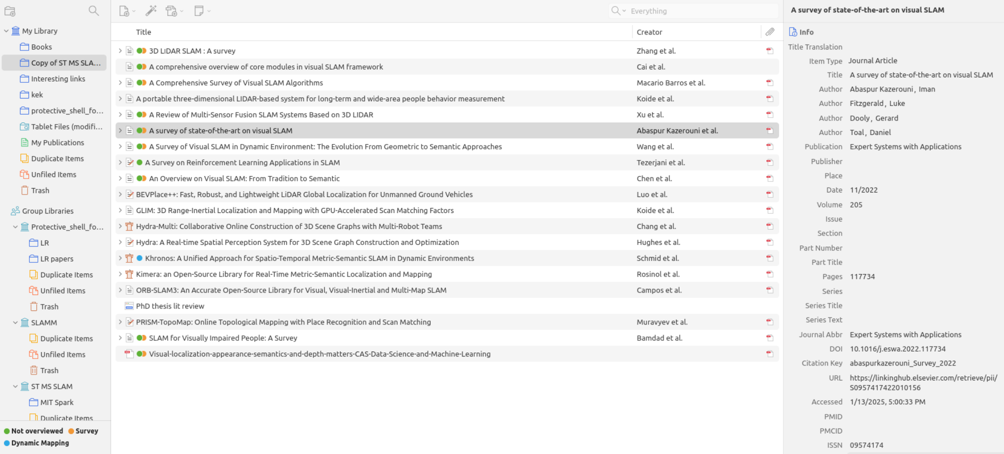
Teamwork & BibTeX Generation

- **Group Libraries:**
  - Share papers, PDFs, and notes with colleagues.
  - Everyone sees the same PDF annotations.
- **BibTeX Workflow:**
  - Right-click Collection → Export Collection → Better BibTeX.
  - Select ”Keep updated” to auto-sync .bib file.
  - **Overleaf:** Link the .bib file via URL or upload manually.
  - **Citation Keys:** Ensure consistency (e.g., [auth][year]).

Obsidian Integration Workflow

- **Goal:** Turn PDF highlights into permanent notes in your ”Second Brain”.
- **Tool:** *Citations* plugin (Obsidian).
- **Process:**
    1. Annotate PDF in Zotero.
    2. Use ”Add Item Note from Annotations”.
    3. Open Obsidian $\longrightarrow$ ”Insert Literature Note”.
    4. Template auto-fills Title, Authors, Abstract, and your Zotero highlights.

Zotero GUI



My Library
Books
Copy of ST MS SLA...
Interesting links
kek
protective_shell_fo...
Tablet Files (modifi...
My Publications
Duplicate Items
Unfiled Items
Trash
Group Libraries
Protective_shell_fo...
LR
LR papers
Duplicate Items
Unfiled Items
Trash
SLAMM
Duplicate Items
Unfiled Items
Trash
ST MS SLAM
MIT Spark
Duplicate Items
Not overviewed
Survey
Dynamic Mapping
Everything
Title
Creator
3D LiDAR SLAM : A survey
Zhang et al.
A comprehensive overview of core modules in visual SLAM framework
Cai et al.
A Comprehensive Survey of Visual SLAM Algorithms
Macario Barros et al.
A portable three-dimensional LIDAR-based system for long-term and wide-area people behavior measurement
Koide et al.
A Review of Multi-Sensor Fusion SLAM Systems Based on 3D LIDAR
Xu et al.
A survey of state-of-the-art on visual SLAM
Abaspur Kazerouni et al.
A Survey of Visual SLAM in Dynamic Environment: The Evolution From Geometric to Semantic Approaches
Wang et al.
A Survey on Reinforcement Learning Applications in SLAM
Tezerjani et al.
An Overview on Visual SLAM: From Tradition to Semantic
Chen et al.
BEVPlace++: Fast, Robust, and Lightweight LiDAR Global Localization for Unmanned Ground Vehicles
Luo et al.
GLIM: 3D Range-Inertial Localization and Mapping with GPU-Accelerated Scan Matching Factors
Koide et al.
Hydra-Multi: Collaborative Online Construction of 3D Scene Graphs with Multi-Robot Teams
Chang et al.
Hydra: A Real-time Spatial Perception System for 3D Scene Graph Construction and Optimization
Hughes et al.
Khronos: A Unified Approach for Spatio-Temporal Metric-Semantic SLAM in Dynamic Environments
Schmid et al.
Kimera: an Open-Source Library for Real-Time Metric-Semantic Localization and Mapping
Rosinol et al.
ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial and Multi-Map SLAM
Campos et al.
PhD thesis lit review
PRISM-TopoMap: Online Topological Mapping with Place Recognition and Scan Matching
Muravyev et al.
SLAM for Visually Impaired People: A Survey
Bamdad et al.
Visual-localization-appearance-semantics-and-depth-matters-CAS-Data-Science-and-Machine-Learning
A survey of state-of-the-art on visual SLAM
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Title Translation
Item Type Journal Article
Title A survey of state-of-the-art on visual SLAM
Author Abaspur Kazerouni, Iman
Author Fitzgerald, Luke
Author Dooly, Gerard
Author Toal, Daniel
Publication Expert Systems with Applications
Publisher
Place
Date 11/2022
Volume 205
Issue
Section
Part Number
Part Title
Pages 117734
Series
Series Title
Series Text
Journal Abbr Expert Systems with Applications
DOI 10.1016/j.eswa.2022.117734
Citation Key abaspurkazerouni_Survey_2022
URL https://linkinghub.elsevier.com/retrieve/pii/S0957417422010156
Accessed 1/13/2025, 5:00:33 PM
PMID
PMCID
ISSN 09574174

Reference Materials (Zotero)

- https://www.zotero.org/
- Zotero SciPDF (Zotero 7)
- Better Notes Plugin
- Better BibTeX Plugin

Reading Strategy with AI

Goal: Process information efficiently without losing depth.

- Using AI assistants (LLMs) to interact with papers.
- Not just summarizing, but querying.

Phase 1: Context & Hallucinations

- **Context Loading (RAG):** Use tools like NotebookLM or custom RAG pipelines to ”chat” with the PDF.
- **Hallucination Check:**
  - Verify specific claims.
  - Ask for page numbers/citations in the response.
  - *”Where exactly does the paper state X?”*

Phase 2: The Truth Seeker

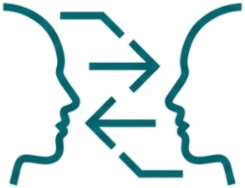
Generate Key Research Questions:

- Instead of ”Summarize this”, ask:
  - ”What is the specific gap this paper fills?”
  - ”What are the limitations explicitly mentioned by authors?”
  - ”Compare method X with baseline Y based on the results tables.”

Phase 3: Deep Analysis

- **Analyze Experiments:**
  - Check experimental setup rigor.
  - Metrics used (and why).
- **Extract Insights:**
  - What worked? What didn't?
  - Why did they choose this architecture?

Secret Technique: Author Debates



Copy This Prompt

Представь, что у нас есть две статьи: Article A и Article B. Ты уже получил их структуру, ключевые идеи и цитаты на основе предыдущего анализа. Сейчас мне нужно, чтобы ты смоделировал научные дебаты между их авторами, строго опираясь на текст статей.

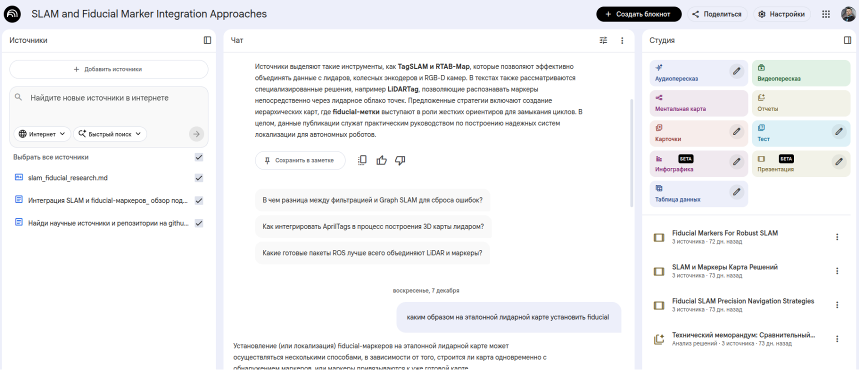
Сначала выступают «авторы» Article A и критически рассматривают Article B. Пусть они укажут, какие аспекты мотивации, постановки задачи, выбора метрик, экспериментального сетапа или интерпретации результатов в Article B вызывают у них сомнения. Просьба формулировать критику в форме реплик, которые могли бы быть произнесены на реальном научном семинаре, и по возможности сопровождать ключевые замечания ссылками на секции, таблицы или рисунки Article B.

Затем дай ответ «авторов» Article B на эту критику. Пусть они объяснят, почему выбрали именно такую формулировку задачи, методику, экспериментальный дизайн или интерпретацию результатов, и укажут, в чём они видят слабые стороны Article A. Также по возможности опирайся на конкретные формулировки и данные из текста Article A.

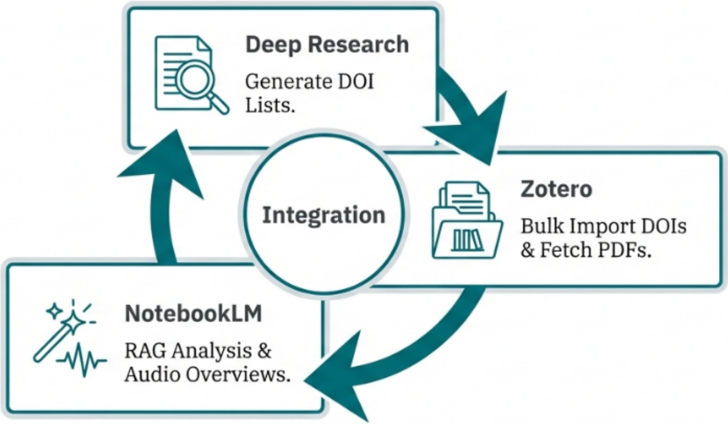
Важно: не добавляй новых фактов, которых нет в статьях. Если какой-то аргумент нельзя обосновать цитатой или явным следствием из текста, обозначь его как гипотетический и минимизируй количество таких допущений.

Why: Forces the model to engage with the logic of the research, not just the text.

NotebookLM as a tool of avoiding hallucinations



Your workflow



Reference Materials

- NotebookLM (Google).

From Plan to Text

Method 1: Plan → Connected Text

- Step 1: Create a detailed hierarchical plan (Chapters, Sections, Bullet points).
- Step 2: Use LLM to expand bullet points into paragraphs.
- **Crucial:** Freeze the structure. Don't let LLM wander.

3 Levels of Text Work

To write clear and structured text, operate on three distinct levels:

1. **Architecture Level:** The overall flow, chapter structure, and logic of the entire document.
2. **Paragraph Level:** The logical unit of thought. Structure within a block.
3. **Sentence Level:** Grammar, style, flow, vocabulary (Polishing).

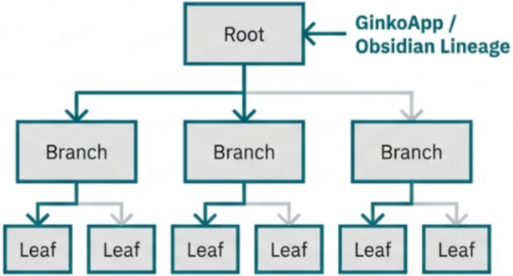
Rules and tools for focusing

The Golden Rule

One Idea = One Paragraph.

If a paragraph has two ideas, split it. If an idea spans two paragraphs, merge it.

Block-Based Writing

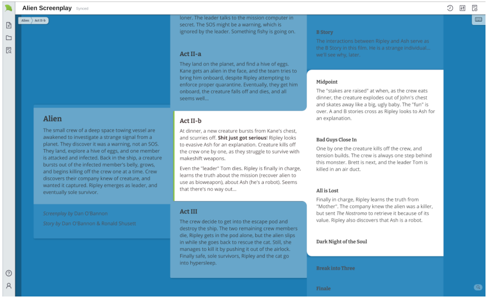


Tools for Block Writing

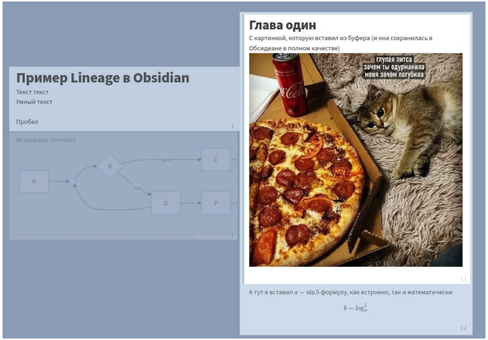
1. **GinkoApp:**
   - Tree-based writing tool.
   - Visualizes the hierarchy of thoughts.
2. **Lineage (Obsidian Plugin):**
   - ”The Gold Standard” for local workflow.
   - Markdown based.
   - Works inside Obsidian.
   - Easy export to LaTeX/Pandoc.

Block Writing in Action

GinkoApp Interface



Lineage (Obsidian)



From Draft to Text

Method 2: Draft → Connected Text

- Step 1: Write a ”Brain Dump” (rough draft, messy English/Russian).
- Step 2: Ask LLM to ”Rewrite in Academic English, maintaining the flow”.
- Better for preserving YOUR original ideas.

Methodological Tricks

- **Reduce Cognitive Load:** Don't ask for "Outcome + Formatting + Style" in one go. Split tasks.
- **Iterative Refinement:**
  - Draft → Critique → Edit.
- **Context Management:** Provide definitions of symbols/terms to keep consistency.

Meta-Prompting (For the Lazy)

- Instead of writing the prompt yourself, ask a stronger model to write it.
- **Workflow:**
  1. Tell DeepSeek/Opus: ”I want to generate a section about X. Write a detailed System Prompt for GPT-4 to do this perfectly.”
  2. Use the generated prompt.

Image Generation for Papers

- **Flux:** Good for diagrams and schematic illustrations.
- **DALL-E 3:** Integrated, follows instructions well, but can be ”cartoony”.
- **Use Case:** Visual Abstracts, concept diagrams.
- **Pro Tip:** Generate parts of diagram and assemble in Figma/TikZ.

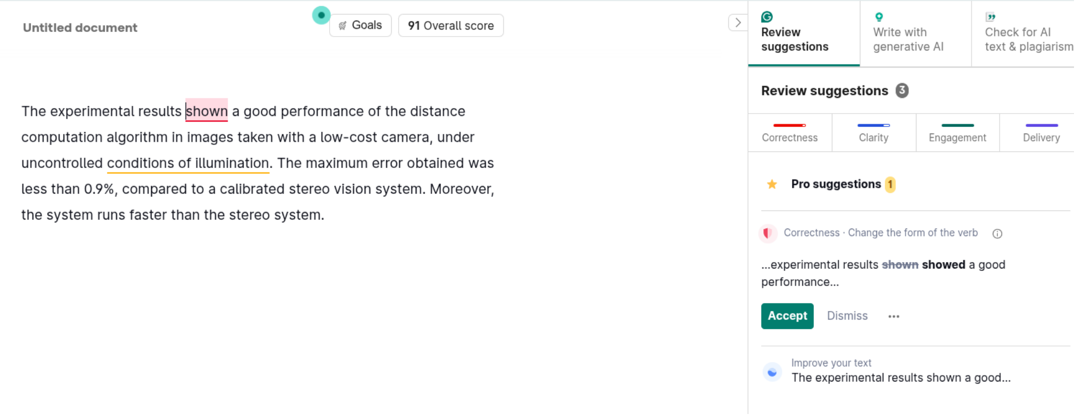
Reference Materials

- GinkoApp
- Obsidian Lineage Plugin
- Using Gingko to Do the Single Method of Academic Writing
- Flux (Black Forest Labs)
- DALL-E 3 (via ChatGPT)

Proofreading: Human vs AI

- **AI Proofreading:** Good for grammar, flow, typo detection.
- **Human Proofreading:** Essential for logic, nuance, and ”vibe”.
- **Strategy:** AI Check → Human Edit → AI Final Polish.
- Use ”Track Changes” mode in prompts (”Show me what you changed”).

Proofreading Interface Example (Grammarly)



Title Engineering

- Title is the marketing of your paper.
- Use AI to generate 20 variations.
- Mix and Match: Combine the best parts.
- Keywords: Ensure SEO-friendly terms are in the title.

Pre-submission Checklist

1. Figures are readable in Black & White?
2. Captions explain the figure fully?
3. Citations are correct?
4. Formatting compliance?
5. Spellcheck done?

AI as Reviewer

Goal: Get feedback BEFORE the actual reviewers destroy you.

- **Tools:**
  - **cspaper:** Specializes in AI conferences (NeurIPS, ICLR).
  - **paperreview.ai**
- Simulate the peer-review process.

Structure of an AI Review

- **Summary:** Does the AI understand your paper? (If not, rewrite).
- **Strengths & Weaknesses:** Crucial feedback.
- **Questions:** What is unclear?
- **Rating:** (e.g., 1-10 scale).
- **Confidence:** (Low confidence = confusing paper).

Warning: Limitations

- **Missing Related Work:** AI often hallucinates papers here. Verification is mandatory.
- **Bias:** Models may prefer certain writing styles.
- **Use it for constructive critique, not for the score.**

Automated Review Interface

← Back

YOUR REVIEW

Paper Length

Pass ✅.

Topic Compatibility

Pass ✅. The paper focuses on Multi-Agent Pathfinding (MAPF) using Imitation Learning and Transformers, which fits well within the "Multiagent Systems" and "Machine Learning" tracks of AAAI.

Minimum Quality

Pass ✅. The paper is complete, technically sound, and presents a clear methodology with extensive experiments.

Prompt Injection and Hidden Manipulation Detection

Pass ✅. No hidden prompts or manipulation detected.

Expected Review Outcome:

Review

The authors propose MAPF-GPT, a decentralized multi-agent pathfinding solver trained via imitation learning on a massive dataset of 1 billion expert observation-action pairs generated by LaCAM. The approach utilizes a transformer-based architecture to predict agent actions from local observations, employing a novel tokenization scheme that encodes map features and agent states. The paper demonstrates that this data-driven approach outperforms state-of-the-art learnable solvers like SCRIMP and DCC in terms of success rate and runtime, while exhibiting zero-shot generalization to unseen map types.

Pros and cons

Pros

- **Significant Dataset Contribution:** The creation and utilization of a 1 billion observation-action pair dataset (**Section 3.2**) is a commendable effort. Generating such a large-scale resource using a strong expert like LaCAM provides a solid foundation for this work and potentially for the broader community if released.
- **Strong Empirical Performance:** The results in **Figure 3** are persuasive. MAPF-GPT-85M consistently outperforms SCRIMP and DCC across 'Random' and 'Mazes' maps. Specifically, on 'Mazes' with 32 agents, MAPF-GPT maintains a success rate above 80% while SCRIMP drops significantly below 60%. The tight confidence intervals (shaded areas) indicate robust performance.

Reference Materials

- https://cspaper.org/
- paperreview.ai

The Detection Problem

- **ICLR Case:** Massive lazy reviewing with AI.
- 21% of reviews were suspected to be AI-generated.
- **Why it matters:** Low effort reviews hurt science.

Signs of Generated Text (Manual)

- **Vocabulary:** Overuse of ”delve”, ”underscore”, ”tapestry”, ”seamless”, ”landscape”.
- **Structure:** Perfect, monotonous paragraph lengths.
- **Lack of Specifics:** Vague claims (”method is robust”) without data (”method achieved 95%”).

Technical Detection

- **Unicode Artifacts:** AI might use weird symbols.
- **Services:**
  - GPTZero, Pangram, Originality.ai.
- **Problem:** High False Positive rate.
- Good papers can be flagged as AI. Don't trust detectors 100%.

Reference Materials

- Pangram Labs
- GPTZero

AI is the accelerator. You are the driver. Keep your hands on the wheel.

Deserve "A" grade!

– Oleg Bulichev

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Цифра, 521